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Activity

1. medical image enhancement

scenario:

An x-ray image has clear bones, but the soft tissues are very dark

- ① Gamma (power-law) transformation ($\gamma < 1$)
- ② It brightens the dark region without affecting the brighter bone areas too much
- ③ Soft tissues become more visible, while bones remain clear, making the x-ray easier to analyze

2. Night Time surveillance

scenario:

A CCTV image is very dark because most pixel values are between 0 and 40

- ① log transformation
- ② log transformation increases the brightness of dark pixels. Negative transformation only inverts image colors and does not improve visibility

- ③ The parking lot becomes brighter, and objects such as vehicles and people are easier to see.

3. Fingerprint Recognition

Scenario:

A fingerprint image has low contrast between ridges and valley

- ① contrast stretching
- ② It increases the differences between ridges and valleys, making fingerprint pattern clear
- ③ The image would become brighter, but the ridge details may not become as clear as with contrast.

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